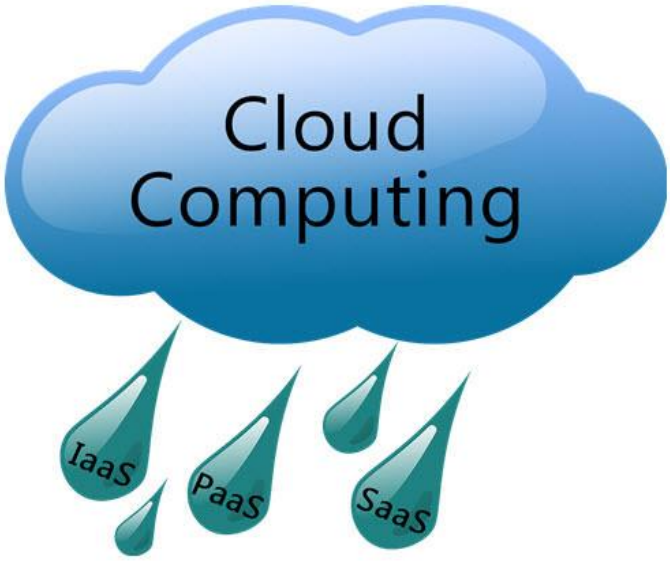


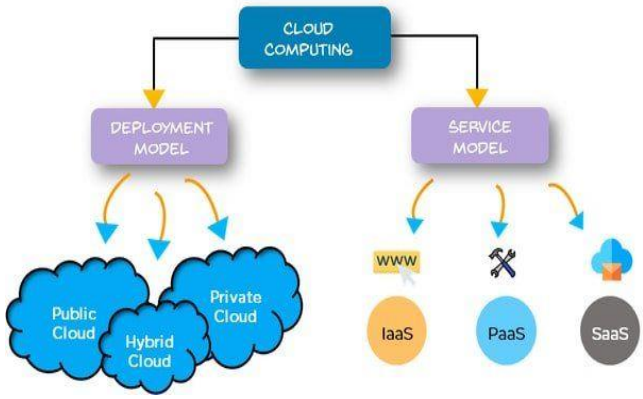
CLOUD COMPUTING COURSE

Lecture4-Cloud Computing Services Models

ABSTRACT
Lecture4 Introduces cloud services models.

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Cloud Computing Models



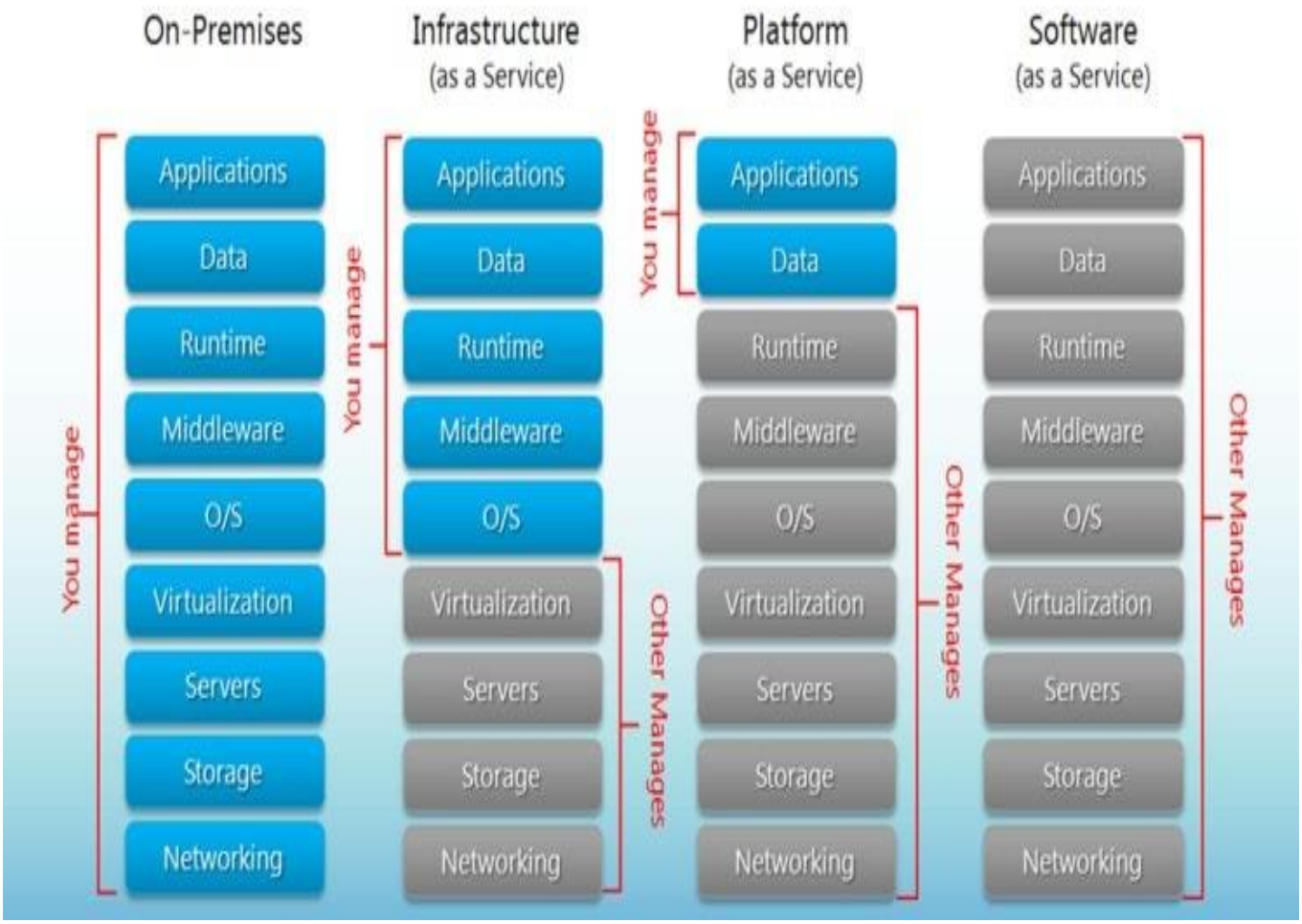
As we learned in the previous lecture that there are two working models for cloud computing:

- Deployment Models
- Service Models

Cloud Computing Service Models

Cloud computing is based on service models. These are categorized into three basic service models, which are:

- Infrastructure-as–a-Service (IaaS)
- Platform-as-a-Service (PaaS)
- Software-as-a-Service (SaaS)



Infrastructure as a service (IaaS): is an instant computing infrastructure, provisioned and managed over the internet. IaaS quickly scales up and down with demand, letting you pay only for what you use. It helps you avoid the expense and complexity of buying and managing your own physical servers and other datacenter infrastructure.

The Management: A cloud computing service provider manages the infrastructure, while you purchase, install, configure, and manage your own software, operating systems, middleware, and applications.

Common IaaS business scenarios

1. **Test and development.** Teams can quickly set up and dismantle test and development environments, bringing new applications to market faster. IaaS makes it quick and economical to scale up dev-test environments up and down.

2. **Website hosting.** Running websites using IaaS can be less expensive than traditional web hosting.

3. **Storage, backup, and recovery.** Organizations avoid the capital outlay for storage and complexity of storage management, which typically requires a skilled staff to manage data and meet legal and compliance requirements. IaaS is useful for handling unpredictable demand and steadily growing storage needs. It can also simplify planning and management of backup and recovery systems.

4. **Web apps.** IaaS provides all the infrastructure to support web apps. Organizations can quickly deploy web apps on IaaS and easily scale infrastructure up and down when demand for the apps is unpredictable.

5. **High-performance computing.** High-performance computing (HPC) helps solve complex problems involving millions of variables or calculations. Examples include earthquake simulations, climate and weather predictions, financial modeling, and evaluating product designs.

Platform as a service (PaaS) is computing model where cloud service providers deliver a set of development kits and programming environments as well as prebuilt application components. These prebuilt components often include back-end frameworks, libraries, database management tools, and other components.

Like IaaS, PaaS includes infrastructure (servers, storage, and networking) but also middleware, development tools, database management systems, and more.

The Management:

PaaS allows you to avoid the expense and complexity of buying and managing software licenses, the underlying application infrastructure and middleware or the development tools and other resources. You manage the applications and services you develop, and the cloud service provider typically manages everything else.

Common PaaS scenarios

Development framework. PaaS provides a framework that developers can build upon to develop or customize cloud-based applications. PaaS lets developers create applications using built-in software components. Cloud features such as scalability, high-availability, and multi-tenant capability are included, reducing the amount of coding that developers must do.

Software as a service (SaaS): allows users to connect to and use cloud-based apps over the Internet. Common examples are email, calendaring, and office tools (such as Microsoft Office 365).

SaaS provides a complete software solution that you purchase on a pay-as-you-go basis from a cloud service provider. You rent the use of an app for your organization, and your users connect to it over the Internet, usually with a web browser. All of the underlying infrastructure, middleware, app software, and app data are located in the service provider's data center.

Management:

The service provider manages the hardware and software, and with the appropriate service agreement, will ensure the availability and the security of the app and your data as well. SaaS allows your organization to get quickly up and running with an app at minimal upfront cost.

Common SaaS scenarios

1. **Free services for personal use**: If you've used a web-based email service such as Outlook, Hotmail, or Yahoo! Mail, then you've already used a form of SaaS. With these services, you log into your account over the Internet, often from a web browser.

2. **For organizational use:** you can rent productivity apps, such as email, collaboration, and calendaring; and sophisticated business applications such as customer relationship management (CRM), enterprise resource planning (ERP), and document management. You pay for the use of these apps by subscription or according to the level of use.

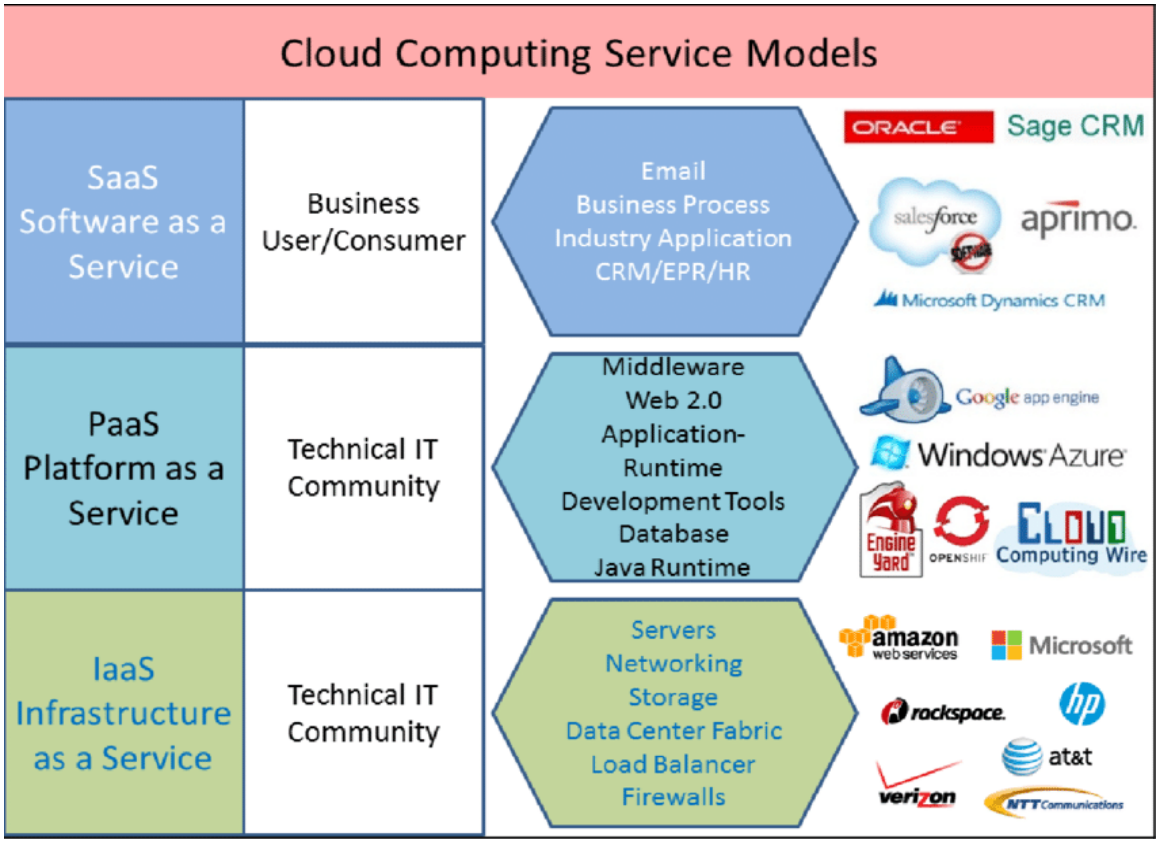
Providers Examples of SaaS, PaaS, and IaaS

Most businesses use a combination of SaaS and IaaS cloud computing service models, and many engage developers to create applications using PaaS, too.

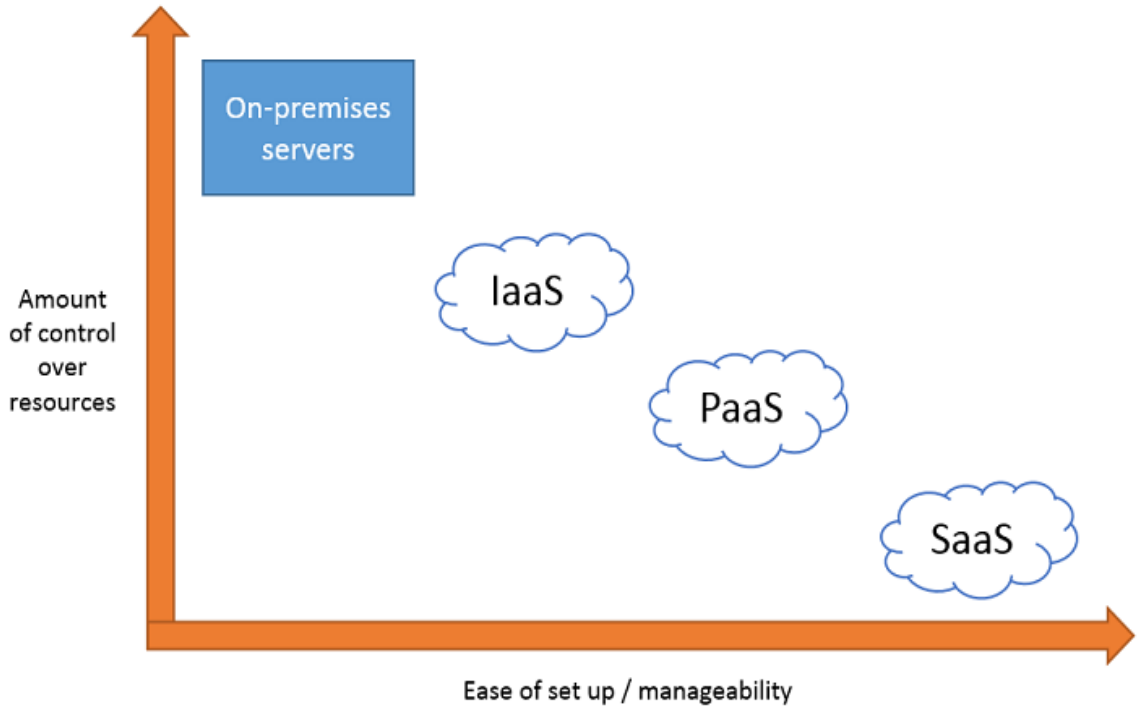
SaaS examples: BigCommerce, Google Apps, Salesforce, Dropbox, MailChimp, ZenDesk, DocuSign, Slack, Hubspot.

PaaS examples: AWS Elastic Beanstalk, Heroku, Windows Azure (mostly used as PaaS), Force.com, OpenShift, Apache Stratos, Magento Commerce Cloud.

IaaS examples: AWS EC2, Rackspace, Google Compute Engine (GCE), Digital Ocean.



Comparison:



Simple Example:

